

Automated OCT-Based Retinal Disease Classification Using Explainable DL

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Abstract

Automated analysis of retinal OCT scans can speed up screening for sight-threatening diseases (e.g., CNV, DME, drusen) and reduce inter-reader variability. This project builds a hybrid, explainable pipeline that (1) trains a CNN classifier on OCT images, (2) uses Grad-CAM to localize suspicious regions, (3) segments ROIs (weakly or with pseudo-masks), (4) extracts interpretable features (area, aspect ratio, mean intensity/color, GLCM texture), and (5) trains a classical classifier (RandomForest) on those features. The hybrid approach combines the pattern-learning power of deep nets with the interpretability and clinical usefulness of feature-based models. Visual explanations (Grad-CAM overlays and ROI masks) plus quantitative feature reports and a feature-based classifier make the system suitable for presentation, validation and clinical discussion.

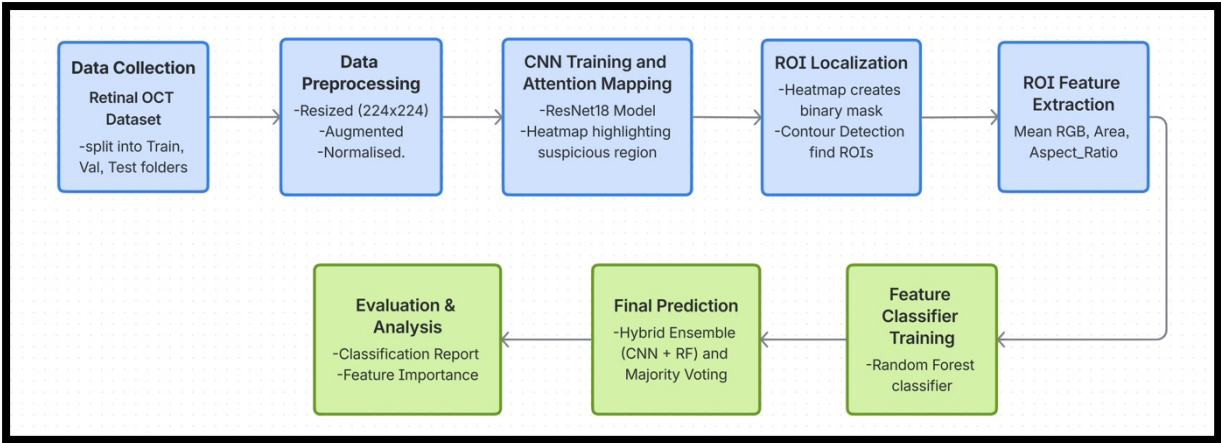
Literature Survey

SNo .	Paper title	Authors — Journal / Year	Problem statement	Methodology (algo / model used)	Metrics used	Limitations
1	OCT-based deep-learning models for identification of retinal pathologies	I. Leandro et al. — <i>Scientific Reports</i> , 2023. (Nature)	Evaluate DL architectures for multi-pathology OCT classification and show explainability.	CNNs (transfer learning), Grad-CAM for interpretability; data preprocessing & augmentation.	Accuracy, AUC, sensitivity, specificity.	Needs large labelled data; limited focus on segmentation; class imbalance discussed. (Nature)
2	RETFound: foundation model for retinal images	Y. Zhou et al. — <i>Nature</i> , 2023. (Nature)	Learn general retinal representations from large unlabeled set for label-efficient downstream tasks.	Self-supervised learning (foundation model), fine-tuning for disease detection.	Downstream classification accuracy, transfer learning gains.	Requires very large pretraining corpus; compute intensive. (Nature)
3	Segmentation-Free OCT-	Z. Chen et al. — <i>PMC</i>	Use 3D OCT volumes	3D CNNs on volume data;	Dice (for segmentation)	Needs volumetric datasets; large

	Volume-Based Deep Learning	(<i>Open Access</i>), 2023. (PMC)	directly for clinical outcome estimation (avoid slice-level segmentation).	end-to-end learning.	tasks when used), classification accuracy.	memory/computation; less interpretability. (PMC)
4	Evaluating DL models for OCT classification	A. Miladinović et al. — <i>Scientific Reports</i> , 2024. (Nature)	Systematically compare architectures under label noise and scarcity.	Multiple CNNs tested, cross-validation, robustness experiments.	Accuracy, robustness measures under noisy labels.	Shows sensitivity to label noise and dataset shifts; calls for explainability. (Nature)
5	Anomaly-guided weakly supervised lesion segmentation in OCT	J. Yang et al. — <i>Elsevier / Computer Vision journal</i> , 2024. (ScienceDirect)	Perform lesion segmentation with only image-level labels using anomaly guidance.	Weakly-supervised segmentation (anomaly maps + guided learning).	Dice, precision/recall on lesion masks.	May produce coarse masks; relies on anomaly detector quality. (ScienceDirect)
6	Text-Driven Weakly Supervised OCT Lesion Segmentation	J. Yang et al. — <i>PubMed / 2024</i> (WSSS approach). (PubMed)	Create pixel-level pseudo labels from image-level labels and textual guidance for OCT lesions.	Weak supervision + structural priors and text guidance.	Dice, IoU, visual quality metrics.	Needs careful text guidance; complexity to implement in practice. (PubMed)
7	U-Net variants comparison for OCT segmentation	J. Kugelman — 2022 (<i>report / thesis</i>). (USC Research)	Compare many U-Net variants on OCT B-scan layer/lesion segmentation.	U-Net, Attention U-Net, Res-U-Net etc.; standardized evaluation.	Dice coefficient, IoU.	Performance varies by dataset; choice of variant depends on task and data. (USC Research)
8	Explainable ensemble methods for OCT detection	J. Yang et al. — <i>PMC 2024</i> (ensemble XAI). (PMC)	Ensemble transfer learning for explainable OCT detection.	Transfer learning, ensembling, Grad-CAM visualizations.	Accuracy, AUC, explanation fidelity.	Ensembles are heavier; explanation still post-hoc (not causal). (PMC)
9	Multi-Fundus / Multi-OCT diseases classification with Grad-CAM	Z. Li et al. — <i>PMC 2023</i> (multi-disease work). (PMC)	Multi-disease classification from fundus/OCT; use Grad-CAM for visualization.	Transfer learning CNNs, Grad-CAM for interpretability.	Accuracy, confusion matrices, visualization examples.	Limited per-disease training examples; standard DL limitations. (PMC)
10	RETFound & label-efficient retinal representation (follow-ups)	Follow-on papers / reviews 2023–24 (self-supervision applied to OCT)	Validate self-supervision for retinal tasks, including OCT classification.	SSL pretraining (SimCLR, DINO variants), fine-tuning.	Transfer accuracy, few-shot performance.	Large-scale pretraining required; still active research. (Nature)

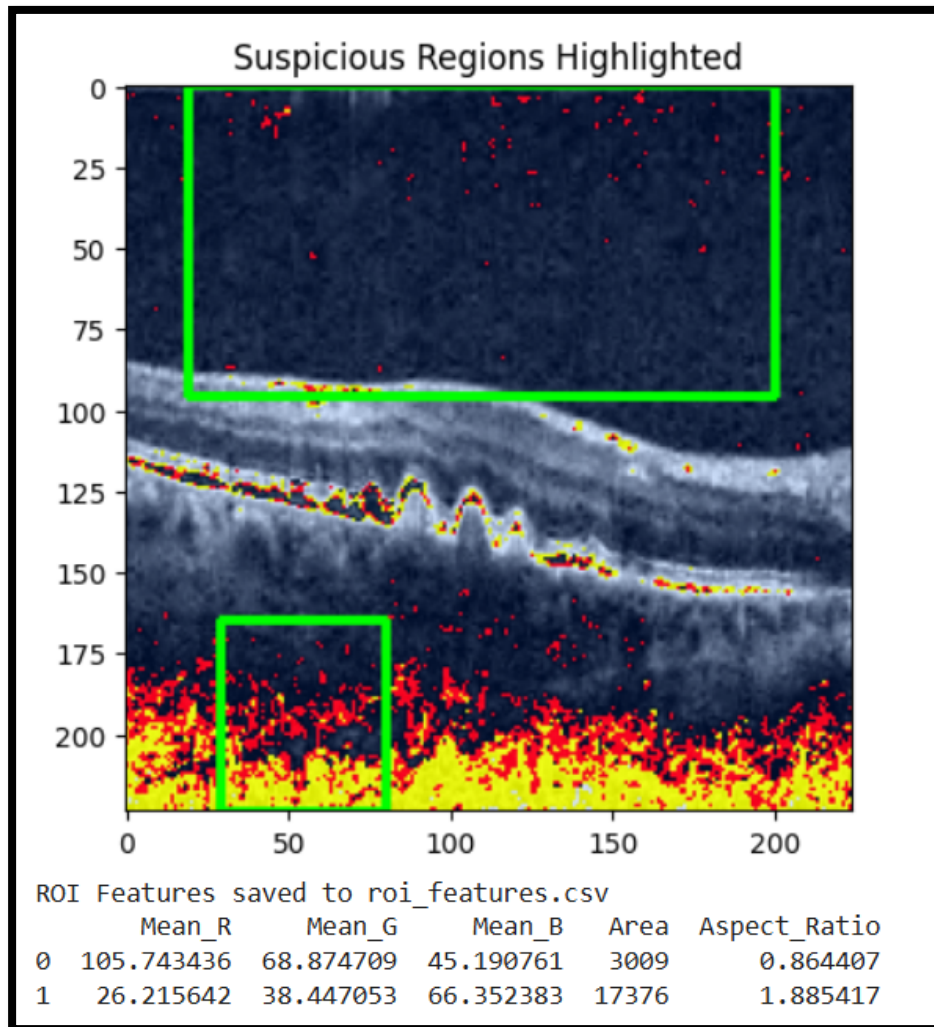
11	DiffuMask / synthetic pseudo-labeling for segmentation	Weijia Wu et al. — (DiffuMask, 2023) — technique used across domains. (ResearchGate)	Use generative models to synthesize pixel-accurate masks for segmentation.	Diffusion models for synthetic masks (DiffuMask).	Segmentation IoU, synthetic realism metrics.	Synthetic realism / domain gap; computational cost. (ResearchGate)
12	RETFound-style foundation and ViT explainable approaches	Vision Transformer + UniMIE + Grad-CAM studies (2024–25) — emergent area. (ResearchGate)	Use ViT or foundation models with explainability for OCT classification.	ViT, image enhancement, Grad-CAM or attention visualization.	Improved accuracy for subtle differences; attention maps.	ViTs need large data and careful explainability alignment. (ResearchGate)

Architectural diagram



Results

Grad-CAM results



Feature extraction CSV preview

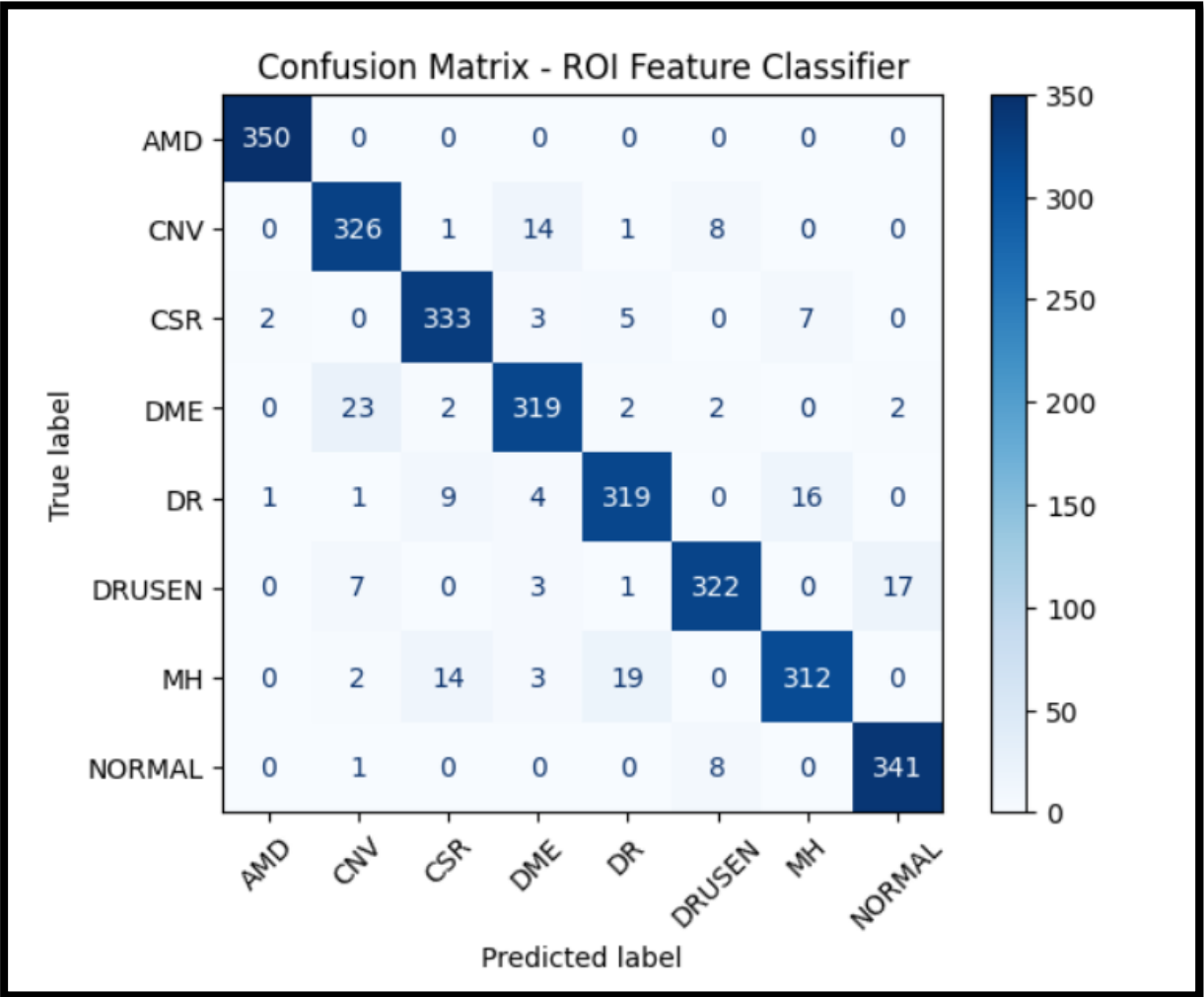
Mean R	Mean G	Mean B	Area	Aspect Ratio
207.9839	109.2373	41.06468	17920	2.8
86.03088	84.79191	108.9257	2720	1.7
80.86292	49.70357	60.61364	11220	2.426471
65.6836	77.24242	99.6416	11476	1.986842
154.6281	96.31332	40.31814	9546	1.743243
55.80195	62.53028	85.42284	13608	1.928571
110.3935	76.16751	51.73152	17748	2.344828
89.6381	99.56583	115.1103	9768	1.261364
184.7491	74.29817	61.84972	11319	1.909091
175.4381	75.22811	32.27103	3472	1.107143
226.8558	51.73643	30.88682	1290	1.433333

Feature-based classifier results

Classification Report (Random Forest on ROI Features):				
	precision	recall	f1-score	support
AMD	0.99	1.00	1.00	350
CNV	0.91	0.93	0.92	350
CSR	0.93	0.95	0.94	350
DME	0.92	0.91	0.92	350
DR	0.92	0.91	0.92	350
DRUSEN	0.95	0.92	0.93	350
MH	0.93	0.89	0.91	350
NORMAL	0.95	0.97	0.96	350
accuracy			0.94	2800
macro avg	0.94	0.94	0.94	2800
weighted avg	0.94	0.94	0.94	2800

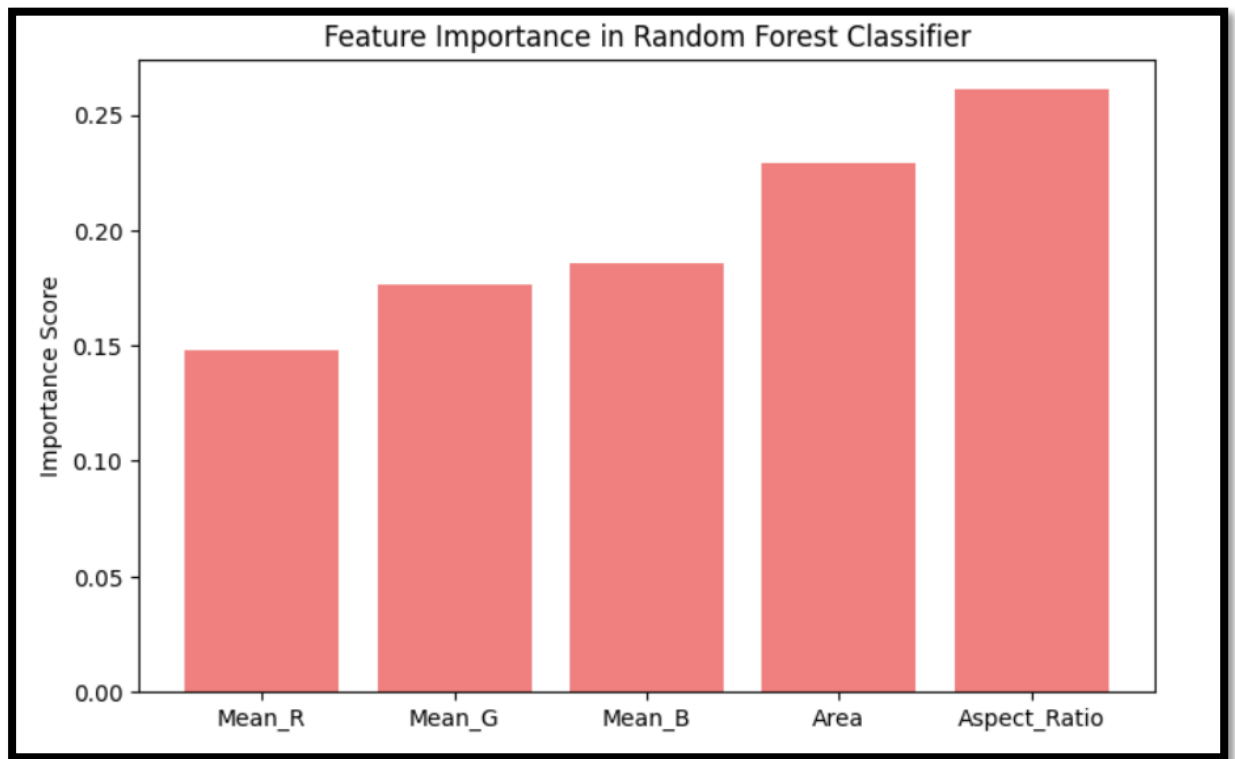
The Random Forest classifier achieved a high overall accuracy of 94% across 8 retinal disease classes using ROI-based features. Precision, recall, and F1-scores were consistently strong, indicating robust and balanced performance, especially for critical conditions like AMD and NORMAL.

Confusion Matrix



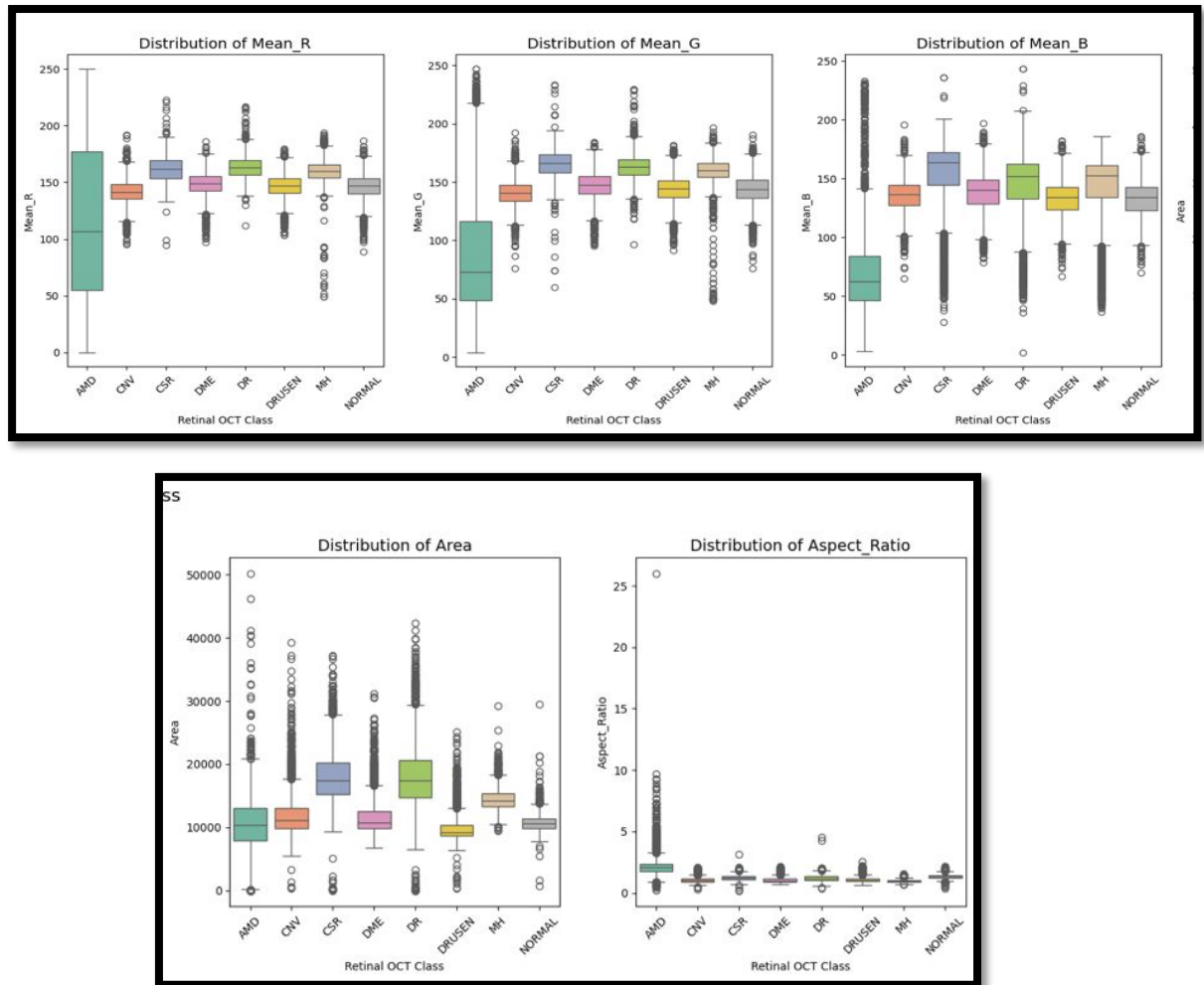
The confusion matrix demonstrates that the feature-based classification model achieves high accuracy, with most classes (e.g., AMD, DR, NORMAL) having a high number of correct predictions (large diagonal values). Minor confusions exist, such as DME samples being occasionally misclassified as CNV (23 cases), but overall, the performance is robust.

Important Features



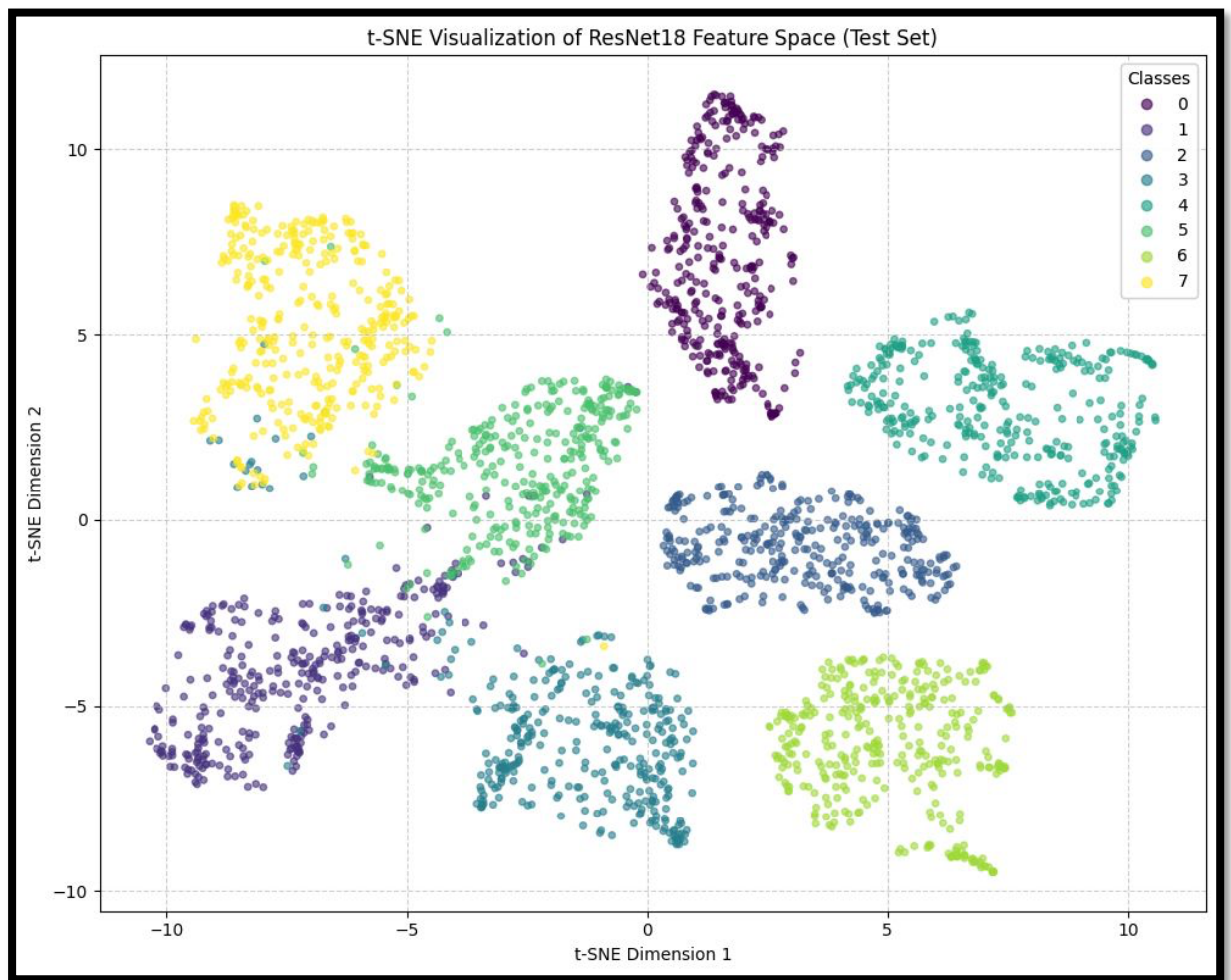
The Feature Importance chart reveals the most decisive factors used by the Random Forest classifier. Specifically, the Aspect_Ratio and the Area of the extracted Region of Interest (ROI) are significantly more important than intensity features (Mean_R/G/B) for distinguishing between the different retinal conditions.

Distribution of Extracted ROI Features by Class



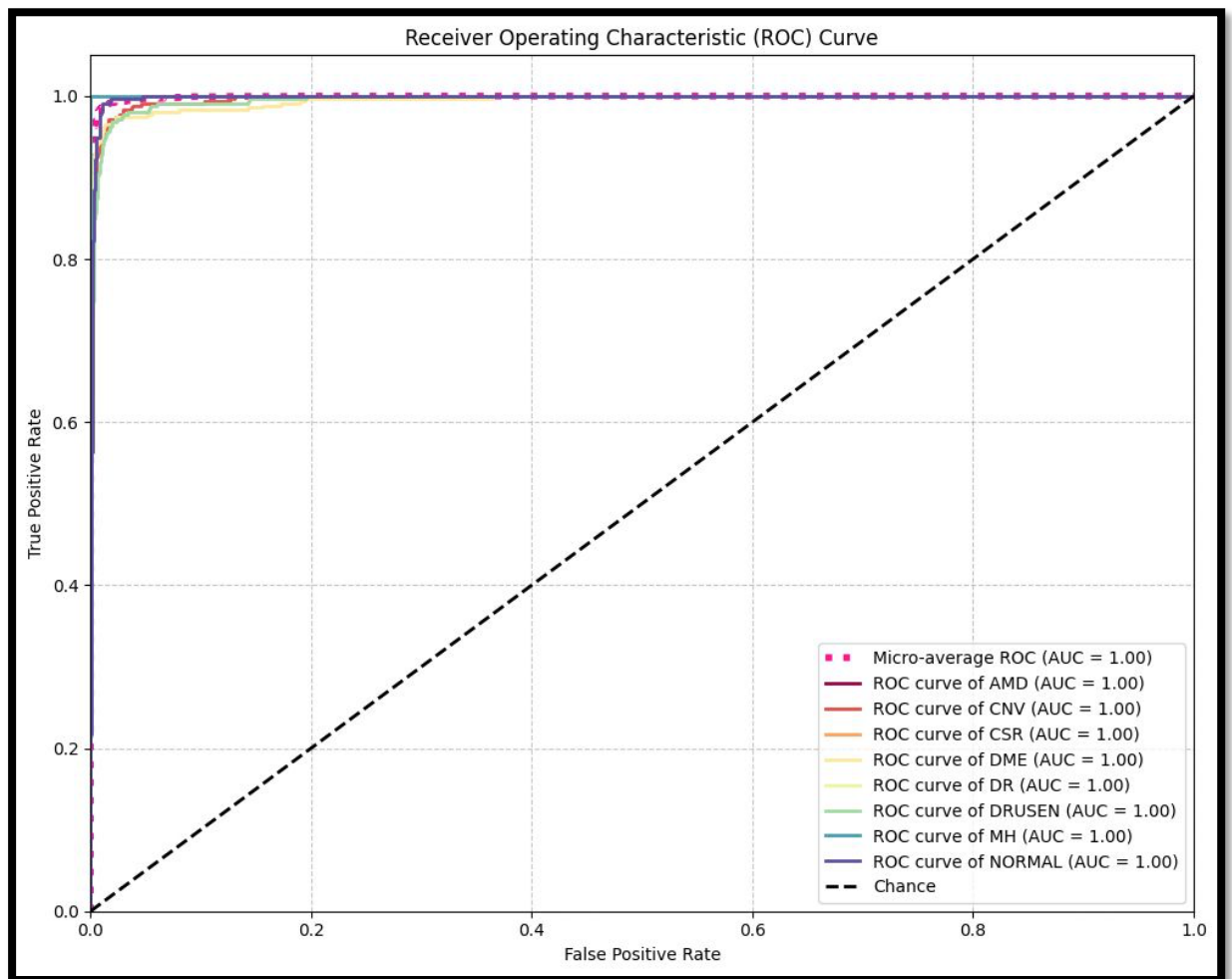
These boxplots display the distribution of extracted region-of-interest (ROI) features such as Mean_R, Mean_G, Mean_B, Area, and Aspect_Ratio across retinal classes. The differences in feature distributions among classes indicate **distinct visual and spatial characteristics** captured from the OCT images, which aid in robust disease differentiation.

t-SNE Visualization of ResNet18 Feature Space (Test Set)

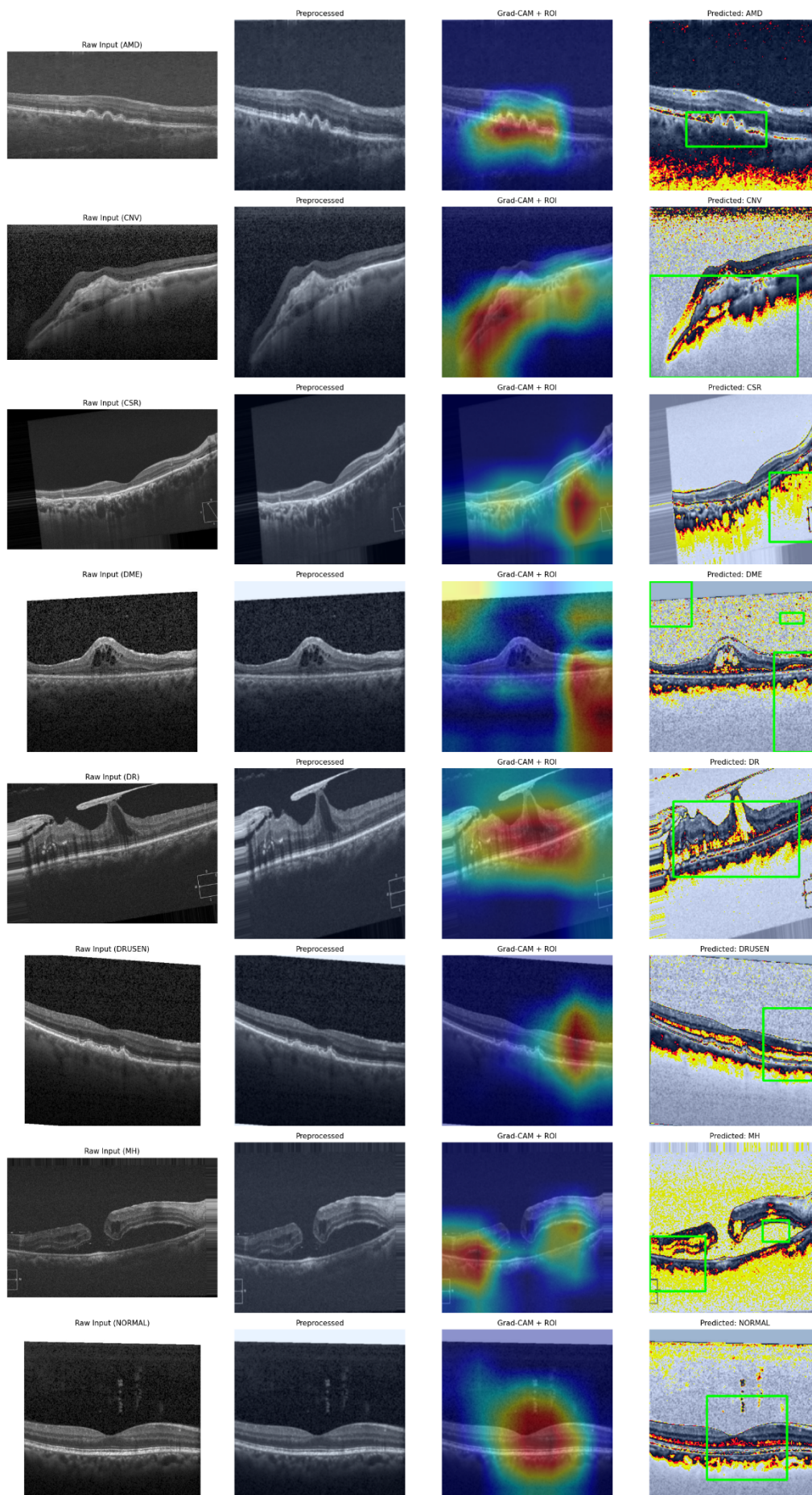


The t-SNE plot visualizes the high-dimensional feature representations extracted by the ResNet18 model. Each color represents a distinct retinal disease class, and the clear clustering with minimal overlap shows that **ResNet18 effectively separates the feature spaces** for different retinal conditions, highlighting its strong representation learning capability.

Receiver Operating Characteristic (ROC) Curve for Multi-Class Retinal Disease Classification



The ROC curve demonstrates that all classes, including AMD, CNV, CSR, DME, DR, DRUSEN, MH, and NORMAL, achieved an AUC (Area Under Curve) of **1.00**, indicating perfect classification performance. The micro-average ROC curve also aligns with this, confirming that the trained model shows **excellent discriminative ability** across all disease categories.



References

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- Li, Z. et al., “Multi-Fundus Diseases Classification Using Retinal Optical Imaging”, *PMC*, 2023. [PMC](#)
- DiffuMask (Weijia Wu et al.), “DiffuMask: Synthesizing images with pixel-level annotations for segmentation”, 2023. [ResearchGate](#)
- Research on Clinically Explainable ViT + UniMIE + Grad-CAM (ResearchGate / arXiv, 2024–25) — examples of ViT & explainability approaches applied to OCT. [ResearchGate+1](#)
- Review resources on U-Net / OCT segmentation and other recent 2024–25 reviews (various journals). [ScienceDirect+1](#)